Does the Yen Carry Unwind Reach Bitcoin?

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A popular claim on X, Reddit and trading forums since the August 2024 sell-off holds that yen unwinds drive crypto sell-offs. It does not hold up in 2018 to 2026 data: once broad risk is accounted for, the yen adds nothing, and a yen-based warning signal did worse than chance.

Recommendation

Do not rely on yen movement as a risk signal for crypto. In my tests, the yen stress carried no information about Bitcoin beyond what equities, volatility, the dollar and US rates already showed, and de-risking on it lowered returns without meaningfully reducing losses. I would watch US 2-year yields and broad risk instead. I hold this view with moderate confidence, pending one final test on Japanese government interventions (H1c, due November 2026). The other 5 key tests showed no meaningful information about Bitcoin .I will update this memo with the H1c result once it is available.

The narrative being tested

Since the global sell-off of 5 August 2024 where Bitcoin briefly fell below US$50,000 , a common market story holds that yen carry unwinds spill into crypto. Investors borrow cheaply in yen to buy higher-yielding or riskier assets, when the yen jumps, those positions are forced to close, and Bitcoin is sold along with everything else.

Initially my thought was that this theory was plausible, and it was further corroborated in 2026 as news of more rate hikes caused fear in the markets: the BOJraised its policy rate to 1.0% in June and 1.25% in September, and Japan intervened to support the yen in April to May and again, jointly with the US, on 31 July. If the consensus theory is true, the yen movements should be an early warning for anyone holding Bitcoin beside equities.

What I found

Before touching the data or doing any tests, I published my hypotheses, data rules and pass criteria, so the results could not be altered or tuned after the tests so the results could not be tuned after seeing them. (details in the appendix).

1. The yen adds nothing once broad risk is accounted for. On stress days, Bitcoin did tend to fall when the yen rose, but the effect was not statistically reliable (one-sided $p = 0.066$ against a 0.05 threshold set in advance). Over the following five days, the link disappeared entirely so a true relationship could not be established.

2. US rates move the yen, not Japanese rates. Day to day, changes in US 2-year yields have a strong relationship with the Yen ($z = -13.5$), while changes in Japanese 2-year yields had negligible effect ($z = 0.49$). This explains a 2026 puzzle: the BOJ kept hiking, yet the yen kept weakening. It also explains why rising Japanese yields did not predict Bitcoin falls.

3. As a warning signal, the yen did worse than chance. A rule that flags sharp, volatile yen rallies fired in 10 episodes from 2023 to September 2026. Yet only 1 was followed by a 10% Bitcoin drawdown within two weeks, against 18.7% for a randomly chosen day.

What it means for a portfolio

Acting on the narrative would have cost money. In a backtest of an 80% global equity, 20% Bitcoin portfolio from 2023 to September 2026, halving the Bitcoin position after each yen signal did worse than doing nothing, worse than timing on VIX, and fell below the 90th percentile of 1,000 random timing schedules (Sharpe 1.17 vs 1.32), as shown in Table 1.

Strategy (Jan 2023 to Sep 2026, after costs)	Annual return	Worst drawdown	Sharpe ratio	Days out of BTC
Buy and hold	28.5%	−17.4%	1.26	0%
De-risk on VIX spikes	27.9%	−17.3%	1.24	1.3%
De-risk on yen signal	25.4%	−16.5%	1.17	21.5%
Random timing, 90th percentile of 1,000 runs	Not computed	Not computed	1.32	21.5% (matched)

Table 1

The yen rule kept the portfolio partly out of Bitcoin on a fifth of all days. That trimmed the worst loss by about one percentage point but gave up about three points of annual return, because a signal that is wrong nine times in ten mostly sits out rallies.

What would change my view

- **Interventions move Bitcoin.** If the final test shows Bitcoin consistently falling after Japanese yen-buying interventions, the channel exists, but only on sudden policy shocks, not as a general stress signal.
- **The link appears after the ETF era settles.** Spot Bitcoin ETFs (January 2024) changed who holds Bitcoin; a clear yen effect in post-2024 data alone would weaken this conclusion.
- **A positioning trigger works.** If yen moves predict Bitcoin only when speculative yen short positions are extreme, the narrative is conditional rather than wrong. I have not tested this yet.

Limitations

- **Limited number of events.** The yen signal fired 10 times in the test period and the VIX signal once, so comparisons between signals rest on very few events.
- **Timing favours the competitor.** The yen signal uses noon New York prices while VIX uses the 16:00 close, giving it four extra hours of information. This makes the yen signal's failure more, not less, conclusive.
- **Limited sample size.** Bitcoin is a relatively new asset and reliable hourly exchange prices, enough liquidity, and overlap with the other series (VIX, FX fixings, CFTC data) only exist from the

mid-2010s. Eight years of data, across very different market regimes, limits how far any result generalises.

- **One test is still open.** The intervention event study (H1c) waits on Japan's Ministry of Finance publishing its July to September 2026 data.

Appendix: results and audit trail

Every pre-registered test completed so far failed its pass rule; pass rules were fixed before any data was analysed.

Test	Question	Key result	Pass rule	Verdict
H1a	Same-day: does yen appreciation coincide with BTC falls on stress days?	$\gamma_0 = -0.58$, one-sided p = 0.066 (2,152 days, 98 stress days)	$\gamma_0 < 0$, $p < 0.05$	Fail
H1b	Does yen appreciation on stress days predict BTC falls over 5 days?	$\gamma_5 = +0.22$, p = 0.60	$\gamma_5 < 0$, $p < 0.05$	Fail
H1d	Do rising Japanese 2Y yields predict BTC falls over 5 days?	$\varphi = +0.0003$ per bp, p = 0.60	$\varphi < 0$, $p < 0.05$	Fail
H2	Does the yen signal flag 10% BTC drawdowns better than VIX?	Precision 10% (1 of 10) vs base rate 18.7%	Beat VIX by 10 pp and beat base rate	Fail
H3	Does de-risking on the yen signal beat VIX and random timing?	See Table 1 above	Beat VIX and 90th pct of random timing	Fail
H1c	Does BTC fall after MoF yen-buying interventions?	[Pending MoF data]	Mean and median CAR < 0, Wilcoxon p < 0.10	Pending

Table 2

Method. Daily data 2018 to September 2026 (training 2018 to 2022, test 2023 to 2026) from FRED, Japan's Ministry of Finance, Yahoo Finance, Bitstamp and the CFTC. Regression control for S&P 500 returns, VIX changes, the dollar excluding the yen, and US 2-year yields, with Newey-West standard errors. H2 and H3 were also run against a second, episode-matched VIX signal; a pass required both.

Audit trail.

Hypotheses and pass rules were frozen in a public, independently timestamped record before any analysis (archived pre-registration); every later change is logged in DEVIATIONS.md. Code, test suite and data sources: github.com/russongit/yen-carry-btc.